Arthi Anbalagan

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SUMMARY

Data Analyst with 4+ years of experience delivering data-driven insights across enterprise environments. Strong expertise in SQL, Python (Pandas), Power BI, Tableau, and advanced Excel for large-scale data analysis, reporting, and automation. Proven track record in building end-to-end ETL pipelines, developing executive dashboards, and enabling business stakeholders to make informed decisions through high-impact analytics. Experienced in working with large structured datasets, performance metrics, and regulatory-style reporting, with a strong focus on data quality, governance, and compliance. Skilled in translating complex business requirements into actionable insights using modern BI and AI-assisted analytics tools.

SKILLS

Data Analytics & Programming: SQL (Advanced), Python (Pandas), DAX, Microsoft Excel (Advanced: Pivot Tables, Power Query)
AI & Advanced Analytics: AI-assisted Data Analysis (ChatGPT, Copilot), Power BI Copilot, Automated Insights
BI, Visualization & Reporting: Power BI, Tableau, IBM Cognos, Interactive Dashboards, Data Visualization
ETL & Data Integration: SSIS, Azure Data Factory (ADF), Power Automate
Data Modeling & KPIs: Data Modeling, KPI Definition, Metric Design, Business Performance Tracking
Data Quality & Governance: Data Validation, Data Quality Assurance, Reconciliation, Anomaly Detection
Business & Stakeholder Skills: Stakeholder Communication, Requirements Gathering, Cross-functional Collaboration
Analytics Execution: Ad-hoc Analysis, Insight Generation, Decision Support, Process Optimization

PROFESSIONAL EXPERIENCE

Data Analyst | CitiusTech, USA **Feb 2025 - Present**

- Analyzed large-scale EHR/EMR and claims datasets from multiple healthcare providers using SQL and Python (Pandas, NumPy) to identify gaps in care, readmission trends, and high-risk patient cohorts across 1M+ patient records.
- Designed and automated ETL pipelines to extract data from HIS/HMS systems, normalize clinical and operational data, and load into a centralized healthcare data warehouse for downstream analytics and reporting.
- Developed interactive executive dashboards in Power BI and Tableau to track HEDIS, CMS quality measures, patient outcomes, and cost metrics, enabling care managers and leadership teams to make data-driven decisions.
- Performed advanced data analysis using Python and SQL to evaluate treatment effectiveness, length of stay, utilization rates, and chronic disease trends, improving population health insights.
- Created automated Excel (advanced formulas, Power Query, pivot models) reports for provider performance benchmarking, reducing manual reporting effort by over 40%.
- Collaborated with clinicians, data engineers, and compliance teams to ensure data accuracy, HIPAA compliance, and proper mapping of ICD-10, CPT, and HL7/FHIR standards across integrated healthcare systems.
- Implemented data validation, reconciliation, and anomaly detection logic to improve data quality across clinical and financial datasets, increasing reporting reliability for regulatory and audit purposes.

Business Intelligence Analyst Intern| Eclinicalworks, India **Jun 2024 - Aug 2024**

- Developed and enhanced 15+ enterprise-level reports using IBM Cognos 11, supporting operational and analytical needs across healthcare teams and improving reporting efficiency by 10%.
- Conducted complex SQL-based data analysis for the 340B compliance program, identifying cost-saving opportunities and validating financial and regulatory data.
- Collaborated with business stakeholders to validate report outputs, ensuring data accuracy, consistency, and alignment with business requirements.
- Automated recurring reporting processes using Power Automate, reducing manual effort by 24% and accelerating report delivery timelines.

Data Analyst| Cognizant Technology Solutions, India **Jul 2019 - Dec 2022**

- Designed and delivered 50+ enterprise Power BI dashboards, scorecards, and operational reports, enabling leadership to track KPIs, trends, and business performance for data-driven decision-making.
- Developed and optimized 100+ complex SQL queries, stored procedures, and views across multiple relational databases, improving ETL performance, data extraction, and report generation time.
- Partnered with business stakeholders and subject matter experts to gather functional requirements, perform data validation, and conduct root cause analysis, delivering accurate and actionable analytical insights.
- Implemented advanced DAX calculations, measures, and time-intelligence functions, improving reporting accuracy and increasing dashboard performance by 28%.
- Built and maintained automated SSIS data pipelines and ETL workflows, enhancing data reliability, ensuring data integrity, and saving approximately 20 hours of manual processing per week.
- Managed multiple concurrent reporting initiatives, supporting Agile delivery cycles, while maintaining high data quality, documentation, and stakeholder satisfaction with on-time delivery.

PROJECTS

Medical Surveillance Plan for Hazardous Medications **Dec 2025**

- Designed a Power BI medical surveillance dashboard for Geisinger to monitor hazardous drug exposure, identifying 5+ risk indicators and proposing a model to cut compliance gaps by 20%.

Medical Insurance Cost Analysis | Python, Pandas, Matplotlib, Seaborn **Nov 2025**

- Analyzed 1,300+ insurance records to identify key cost factors and built a regression model with 85% accuracy to predict charges, enabling data-driven insights for cost optimization.

Loan Approval Decision-making Using Datamining Models **Jun 2025**

- Led a project utilizing IBM SPSS and StatTools for advanced Regression models, achieved 90% accuracy through data preprocessing and reducing loan approval processing time by 20%.

Development of an In-patient Surgeries DBMS for UPMC **Jan 2025**

- Developed and implemented a centralized database using SQL to manage data for over 10,000 patients at UPMC's in-patient surgeries and streamlined data access, reducing administrative workload by 30%.

EDUCATION

Master of Science in Information Systems| Pennsylvania State University, PA **Jan 2023 - Dec 2024**
Bachelor of Engineering | Anna University, Chennai, India **Jul 2015 - May 2019**